



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 2022

Course: B. Pharmacy

Sem: VI

Subject Name: Pharmaceutical Quality Assurance

Subject Code: BP606T

Max. Marks: 75

Date: 28/07/2022

Duration: 3.45 Hrs.

Instructions –

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

Q. 1. Multiple Choice Questions (MCQs) = 20 x 1 = 20 (All the questions are compulsory)

- 1) NABL accreditation has a validity period of _____ years.
a) 1 b) 2 c) 3 d) 4
- 2) The is a process of assurance that the specific system, premises or equipment are able to achieve the predetermined acceptance criteria to confirm the attributes.
a) Validation master plan b) Calibration
b) Evaluation d) Qualification
- 3) ISO Certification is valid for _____ years.
a) 1 b) 2 c) 3 d) 4
- 4) Which guideline stands for evaluation of stability data?
a) Q1B b) Q1C c) Q1D d) Q1E
- 5) P-D-C-A stands for.....
a) Proceed-Do-Check-Act b) Plan-Do-Check-Act
c) Plan-Do-Correct-Act d) Prepare-Do-Check-Act
- 6) Physical dimension of equipment and accessories comes under which qualification?
a) Operational qualification (OQ) b) Installation qualification (IQ)
c) Performance qualification (PQ) d) Design qualification (DQ)
- 7) _____ airlock is mainly used in Parenteral manufacturing areas.

- a) Cascade airlock b) Sink airlock
c) Bubble airlock d) Hermetical airlock

8) For Hydrolytic resistance test, at least ____ number of containers to be used for the 5 ml capacity glass containers.
a) 20 b) 10 c) 5 d) 3

9) Fragmentation test applicable to butyl rubber closures, the total number of fragments is NMT _____.
a) 5 b) 10 c) 15 d) 20

10) As per USFDA, Microbiological limit for clean area class 1000 is _____ cfu/ft³.
a) <1 b) <2 c) <3 d) <5

11) Non-clinical laboratory study records must be retained for at least ____ years, when application for research is approved by FDA.
a) 1 b) 2 c) 3 d) 5

12) _____ validation is performed on already completed process.
a) Prospective b) Concurrent c) Retrospective d) Qualification

13) Non-clinical laboratory study protocol is signed and approved by _____.
a) Study Director b) Sponsor c) QA Unit d) Both a & b

14) ____ is degree of closeness of a series of measurement obtained using multiple samples of same substance.
a) Accuracy b) Precision c) Specificity d) Linearity

15) Waste which is potentially dangerous to human health is called as.....
a) Non-hazardous waste b) Hazardous waste
c) Inert substances waste d) Biomedical waste

16) ____ means an action taken to resolve a problem with therapeutic goods having deficiencies in quality, efficacy or safety.
a) Qualification b) Drug recall c) Quality review d) Complaint

17) As per CDSCO, Class II Drug recall should be completed within _____.
a) 3 days b) 10 days c) 30 days d) 7 days

18) _____ audit is performed to assess the GMP & cGMP compliance.
a) External audit b) Internal audit c) Regulatory audit d) Operational audit

19) Product quality complaint should be investigated within _____ days.

a) 5

b) 3

c) 10

d) 7

20) Only complaint sample shows OOS result, due to improper storage and retained sample

meets specifications, this complaint is categorized as _____

a) Confirmed complaint

b) Non-confirmed complaint

c) Counter complaint

d) Tamper complaint

Q. 2. Long Answers = 2 x 10 = 20 (Answer 2 out of 3)

1. Describe process of harmonization and elaborate the ICH stability testing guidelines.
2. Write in detail about Analytical Method Validation. Give process of Calibration of pH meter.
3. Explain the quality control tests for glass containers and rubber closures.

Q. 3. Short Answers = 7 x 5 = 35 (Answer 7 out of 9)

1. Explain the process of Complaint Handling.
2. Describe Elements of QbD.
3. Explain principles and procedure for NABL accreditation.
4. Write a note on Training and Personnel health and hygiene. Explain in short Maintenance of sterile areas.
5. Give an account of purchase specifications and maintenance of stores for raw materials.
6. Discuss the Protocol for Conduct of a Nonclinical Laboratory Study and Records and Reports per GLP.
7. Write a note on Quality Audit and Master Formula Record.
8. Differentiate between QA & QC. Enlist the steps for registration for ISO 9000.
9. Write a short note on Good Warehousing Practice.

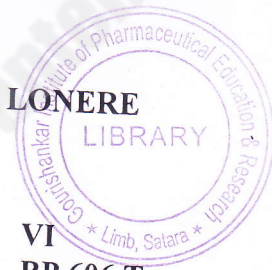
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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
End Semester Examination – Winter 2022

Date: 20/02/2023

Course : B. Pharmacy
Subject Name : Quality Assurance
Max Marks : 75

Sem: VI
Subject Code : BP 606 T
Duration : 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Differentiate between QA and QC.
- ii) What are purchase specifications for equipment?
- iii) Define validation and calibration.
- iv) What is master formula record? Give example.
- v) Enlist philosophies of TQM. Explain any one.
- vi) Explain different types of process validation.
- vii) Explain hydrolytic resistance test for glass containers.
- viii) Enlist steps for ISO registration.
- ix) Define Recall and classify it.
- x) What is SOP and quality audit?

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) Discuss in detail Analytical Method Validation.
- ii) Give scope for NABL accreditation and discuss in detail procedure for it.
- iii) Give purpose of ICH guidelines and discuss ICH stability testing guideline

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Write a note on maintenance of sterile area and control of contamination.
- ii) Describe complaint evaluation.
- iii) Discuss in brief different elements of GLP.
- iv) Describe quality control tests for rubber.
- v) Write a note on Good Warehousing Practice.
- vi) Explain elements of QbD.
- vii) Discuss qualification of UV-Visible Spectrophotometer.
- viii) Write a note on personnel training, health and hygiene. Give an account of their responsibilities.
- ix) Discuss equipment qualification.

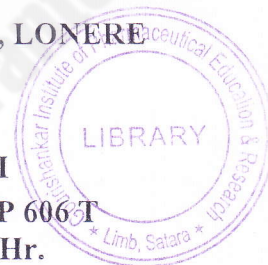
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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
End Semester Examination – Winter 2022

Date: 20/02/2023

Course : B. Pharmacy
Subject Name : Quality Assurance
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Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Enlist benefits of ISO14000.
- ii) Define NABL accreditation.
- iii) Give the difference between QA and QC.
- iv) Enlist Q series of ICH Guidelines.
- v) What is contamination and cross contamination?
- vi) Define TQM. Discuss its advantages.
- vii) What is Batch Formula Record?
- viii) Classify packaging material used in the pharmaceutical industry with example.
- ix) Define Qualification and Calibration.
- x) What is warehousing and material management?

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) Discuss the importance of Good Laboratory Practices. Explain protocol content of Non-clinical laboratory study.
- ii) What is Validation? Explain types of validation and discuss validation master plan.
- iii) Define Quality control. Explain quality control test for containers.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Define airlock. Enlist different types of airlock and explain their working.
- ii) Discuss in brief different elements of GLP.
- iii) Discuss the QSEM guidelines as per ICH.
- iv) Write about the concept of quality by design (QbD) in pharma industry.
- v) Write a note on Master Formula Record.
- vi) Discuss qualification of UV-Visible Spectrophotometer.
- vii) Explain steps involved in complaint handling.
- viii) Explain various documents maintained by quality control department.
- ix) Explain about personnel responsibilities, training and hygiene.

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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Supplementary Examination - Winter 2023

Date: 10/01/2024

Course: B. Pharmacy
Subject Name: Quality Assurance
Max Marks : 75

Sem : VI
Subject Code : BP606T
Duration : 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams/figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Give difference between QA and QC.
- ii) Enlist Q series of ICH guidelines.
- iii) Write down the importance of documentation in pharma industry.
- iv) Define NABL acceleration.
- v) What is mean by warehousing
- vi) What is SOP?
- vii) Enlist benefits of ISO 9000.
- viii) What is mean by complaints and recalling?
- ix) Classify packing material with examples.
- x) Define GMP and give its importance.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) What is mean by BFR and MFR? Explain them in detail.
- ii) Write the principles of Calibration, Validation and Qualification.
- iii) How to maintain organization and personal in pharmaceutical industry.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Explain in detail Quality control test for containers.
- ii) Which are the various elements of TQM?
- iii) Explain QSEM guidelines as per ICH.
- iv) Explain steps involved in complaint handling.

- v) What is mean by premises in the consideration of control of contamination in pharmaceutical industry?
- vi) Discuss in brief different elements of GLP.
- vii) Write a note on QbD.
- viii) Mention the various steps for registration of ISO 9000 and 14000.
- ix) Give the general principles of Analytical Method Validation.

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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE		
Regular/Supplementary Summer Examination – 2024		
Course: B. Pharmacy	Semester: VI	
Subject Name: Quality Assurance	Subject Code: BP606T	
Max Marks: 75	Date: 26-06-2024	Duration: 3 Hr.
Instructions to the Students:		
1. All questions are compulsory		
2. Draw diagrams / figures wherever necessary		
3. Figures to right indicate full marks		
Q.1.	Objective Type Questions (Answer All the Questions)	(10X2) =20
i)	Write the importance of SOP in manufacturing.	
ii)	Enlist the significances of Batch Formula Record.	
iii)	Write about Waste Disposal.	
iv)	State the importance of Contamination Control.	
v)	Write the general provisions of GLP.	
vi)	Write brief on 'Quality Audit'.	
vii)	Give details on Handling of Return Good.	
viii)	Brief note on concept of GMP.	
ix)	State the importance of Records and Reports.	
x)	What is ISO 9000 and ISO 14000.	
Q.2.	Long Answers (Answer 2 out of 3)	(10X2) =20
i)	Explain the ICH Stability Testing Guidelines.	
ii)	Discuss in details about GLP in Pharmaceuticals Industry.	
iii)	Explain the quality controls tests for Containers and Closures.	
Q.3.	Short Answers (Answer 7 out of 9)	(5 X 7) = 35
i)	Write a note on calibration of pH meter.	
ii)	What are the Complaints? Add a note on evaluation of Complaints.	
iii)	Discuss about maintenance of Equipments.	
iv)	Compare the concept of QA and QC as per GMP.	
v)	Purchase specification and maintenance of Stores for Raw Materials.	
vi)	Note on Plant Layout of Pharmaceutical industry.	
vii)	Compare the concept of Qualification and Validation.	
viii)	Write brief on 'Secondary Packing Materials'.	
ix)	Discuss briefly the Sterile Area's importance & its maintenance.	
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